Machine Learning Algorithms Laboratory
Experiment 2
Email Spam/Ham Classification using Naïve Bayes and KNN

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Aim

To classify emails as spam or ham using Naïve Bayes and K-Nearest Neighbors (KNN) algorithms and compare their performance.

Objective

- Implement Naïve Bayes and KNN classifiers.
- Compare Gaussian, Multinomial and Bernoulli Naïve Bayes.
- Study the effect of different values of k in KNN.
- Tune the KNN model using GridSearchCV and RandomizedSearchCV.
- Compare the performance of the classifiers.

Theory

Naïve Bayes is a supervised learning algorithm based on Bayes' theorem. It assumes that the features are independent of each other. It is widely used for text classification problems like spam detection.

K-Nearest Neighbors (KNN) is a supervised learning algorithm that classifies a new sample by looking at its nearest neighbours. The prediction depends on the value of k, which represents the number of neighbours considered.

Dataset

The Spambase dataset from Kaggle was used in this experiment. It contains email samples with numerical features and a target column that classifies each email as spam or ham.

EDA

The following steps were performed during exploratory data analysis.

- Loaded the dataset.
- Checked the shape and information of the dataset.
- Verified missing values.
- Displayed summary statistics.
- Plotted class distribution.
- Generated histograms, boxplots and correlation heatmap.
- Normalized the features before training the models.

```
import time

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.model_selection import GridSearchCV
from sklearn.model_selection import RandomizedSearchCV
from sklearn.model_selection import cross_val_score

from sklearn.preprocessing import MinMaxScaler

from sklearn.naive_bayes import GaussianNB
from sklearn.naive_bayes import MultinomialNB
from sklearn.naive_bayes import BernoulliNB

from sklearn.neighbors import KNeighborsClassifier

from sklearn.metrics import accuracy_score
from sklearn.metrics import precision_score
from sklearn.metrics import recall_score
from sklearn.metrics import f1_score
from sklearn.metrics import roc_auc_score
from sklearn.metrics import confusion_matrix
from sklearn.metrics import ConfusionMatrixDisplay
from sklearn.metrics import RocCurveDisplay
from sklearn.metrics import PrecisionRecallDisplay

df = pd.read_csv("spambase_csv.csv")

print("Dataset Loaded Successfully")
```

```
Dataset Loaded Successfully
```

```
df = pd.read_csv("spambase_csv.csv")

print("Dataset Loaded Successfully")
```

```
Dataset Loaded Successfully
```

```
df.head()
```

```
   word_freq_make  word_freq_address  word_freq_all  word_freq_3d  \
0            0.00               0.64           0.64           0.0
1            0.21               0.28           0.50           0.0
2            0.06               0.00           0.71           0.0
3            0.00               0.00           0.00           0.0
4            0.00               0.00           0.00           0.0

   word_freq_our  word_freq_over  word_freq_remove  word_freq_internet
```

```
\
0           0.32              0.00               0.00                  0.00

1           0.14              0.28               0.21                  0.07

2           1.23              0.19               0.19                  0.12

3           0.63              0.00               0.31                  0.63

4           0.63              0.00               0.31                  0.63


   word_freq_order  word_freq_mail  ...  char_freq_%3B  char_freq_%28
\
0             0.00            0.00  ...           0.00          0.000

1             0.00            0.94  ...           0.00          0.132

2             0.64            0.25  ...           0.01          0.143

3             0.31            0.63  ...           0.00          0.137

4             0.31            0.63  ...           0.00          0.135


   char_freq_%5B  char_freq_%21  char_freq_%24  char_freq_%23  \
0            0.0          0.778          0.000          0.000
1            0.0          0.372          0.180          0.048
2            0.0          0.276          0.184          0.010
3            0.0          0.137          0.000          0.000
4            0.0          0.135          0.000          0.000

   capital_run_length_average  capital_run_length_longest  \
0                       3.756                          61
1                       5.114                         101
2                       9.821                         485
3                       3.537                          40
4                       3.537                          40

   capital_run_length_total  class
0                       278      1
1                      1028      1
2                      2259      1
3                       191      1
4                       191      1

[5 rows x 58 columns]
```

```
print("Shape :", df.shape)

print("\nColumns")
```

```
print(df.columns)

print("\nData Types")
print(df.dtypes)

print("\nInformation")
df.info()
```

```
Shape : (4601, 58)

Columns
Index(['word_freq_make', 'word_freq_address', 'word_freq_all',
'word_freq_3d',
       'word_freq_our', 'word_freq_over', 'word_freq_remove',
       'word_freq_internet', 'word_freq_order', 'word_freq_mail',
       'word_freq_receive', 'word_freq_will', 'word_freq_people',
       'word_freq_report', 'word_freq_addresses', 'word_freq_free',
       'word_freq_business', 'word_freq_email', 'word_freq_you',
       'word_freq_credit', 'word_freq_your', 'word_freq_font',
'word_freq_000',
       'word_freq_money', 'word_freq_hp', 'word_freq_hpl',
'word_freq_george',
       'word_freq_650', 'word_freq_lab', 'word_freq_labs',
'word_freq_telnet',
       'word_freq_857', 'word_freq_data', 'word_freq_415',
'word_freq_85',
       'word_freq_technology', 'word_freq_1999', 'word_freq_parts',
       'word_freq_pm', 'word_freq_direct', 'word_freq_cs',
'word_freq_meeting',
       'word_freq_original', 'word_freq_project', 'word_freq_re',
       'word_freq_edu', 'word_freq_table', 'word_freq_conference',
       'char_freq_%3B', 'char_freq_%28', 'char_freq_%5B', 'char_freq_
%21',
       'char_freq_%24', 'char_freq_%23', 'capital_run_length_average',
       'capital_run_length_longest', 'capital_run_length_total',
'class'],
      dtype='str')

Data Types
word_freq_make                 float64
word_freq_address              float64
word_freq_all                  float64
word_freq_3d                   float64
word_freq_our                  float64
word_freq_over                 float64
word_freq_remove               float64
word_freq_internet             float64
word_freq_order                float64
word_freq_mail                 float64
word_freq_receive              float64
```

```
word_freq_will                float64
word_freq_people              float64
word_freq_report              float64
word_freq_addresses           float64
word_freq_free                float64
word_freq_business            float64
word_freq_email               float64
word_freq_you                 float64
word_freq_credit              float64
word_freq_your                float64
word_freq_font                float64
word_freq_000                 float64
word_freq_money               float64
word_freq_hp                  float64
word_freq_hpl                 float64
word_freq_george              float64
word_freq_650                 float64
word_freq_lab                 float64
word_freq_labs                float64
word_freq_telnet              float64
word_freq_857                 float64
word_freq_data                float64
word_freq_415                 float64
word_freq_85                  float64
word_freq_technology          float64
word_freq_1999                float64
word_freq_parts               float64
word_freq_pm                  float64
word_freq_direct              float64
word_freq_cs                  float64
word_freq_meeting             float64
word_freq_original            float64
word_freq_project             float64
word_freq_re                  float64
word_freq_edu                 float64
word_freq_table               float64
word_freq_conference          float64
char_freq_%3B                 float64
char_freq_%28                 float64
char_freq_%5B                 float64
char_freq_%21                 float64
char_freq_%24                 float64
char_freq_%23                 float64
capital_run_length_average    float64
capital_run_length_longest      int64
capital_run_length_total        int64
class                           int64
dtype: object
```

```
Information
<class 'pandas.DataFrame'>
RangeIndex: 4601 entries, 0 to 4600
Data columns (total 58 columns):
 #   Column                      Non-Null Count  Dtype
---  ------                      --------------  -----
 0   word_freq_make              4601 non-null   float64
 1   word_freq_address           4601 non-null   float64
 2   word_freq_all               4601 non-null   float64
 3   word_freq_3d                4601 non-null   float64
 4   word_freq_our               4601 non-null   float64
 5   word_freq_over              4601 non-null   float64
 6   word_freq_remove            4601 non-null   float64
 7   word_freq_internet          4601 non-null   float64
 8   word_freq_order             4601 non-null   float64
 9   word_freq_mail              4601 non-null   float64
 10  word_freq_receive           4601 non-null   float64
 11  word_freq_will              4601 non-null   float64
 12  word_freq_people            4601 non-null   float64
 13  word_freq_report            4601 non-null   float64
 14  word_freq_addresses         4601 non-null   float64
 15  word_freq_free              4601 non-null   float64
 16  word_freq_business          4601 non-null   float64
 17  word_freq_email             4601 non-null   float64
 18  word_freq_you               4601 non-null   float64
 19  word_freq_credit            4601 non-null   float64
 20  word_freq_your              4601 non-null   float64
 21  word_freq_font              4601 non-null   float64
 22  word_freq_000               4601 non-null   float64
 23  word_freq_money             4601 non-null   float64
 24  word_freq_hp                4601 non-null   float64
 25  word_freq_hpl               4601 non-null   float64
 26  word_freq_george            4601 non-null   float64
 27  word_freq_650               4601 non-null   float64
 28  word_freq_lab               4601 non-null   float64
 29  word_freq_labs              4601 non-null   float64
 30  word_freq_telnet            4601 non-null   float64
 31  word_freq_857               4601 non-null   float64
 32  word_freq_data              4601 non-null   float64
 33  word_freq_415               4601 non-null   float64
 34  word_freq_85                4601 non-null   float64
 35  word_freq_technology        4601 non-null   float64
 36  word_freq_1999              4601 non-null   float64
 37  word_freq_parts             4601 non-null   float64
 38  word_freq_pm                4601 non-null   float64
 39  word_freq_direct            4601 non-null   float64
 40  word_freq_cs                4601 non-null   float64
 41  word_freq_meeting           4601 non-null   float64
 42  word_freq_original          4601 non-null   float64
```

```
 43  word_freq_project           4601 non-null   float64
 44  word_freq_re                4601 non-null   float64
 45  word_freq_edu               4601 non-null   float64
 46  word_freq_table             4601 non-null   float64
 47  word_freq_conference        4601 non-null   float64
 48  char_freq_%3B               4601 non-null   float64
 49  char_freq_%28               4601 non-null   float64
 50  char_freq_%5B               4601 non-null   float64
 51  char_freq_%21               4601 non-null   float64
 52  char_freq_%24               4601 non-null   float64
 53  char_freq_%23               4601 non-null   float64
 54  capital_run_length_average  4601 non-null   float64
 55  capital_run_length_longest  4601 non-null   int64
 56  capital_run_length_total    4601 non-null   int64
 57  class                       4601 non-null   int64
dtypes: float64(55), int64(3)
memory usage: 2.0 MB

df.describe()

       word_freq_make  word_freq_address  word_freq_all  word_freq_3d
\
count     4601.000000        4601.000000    4601.000000   4601.000000

mean         0.104553           0.213015       0.280656      0.065425

std          0.305358           1.290575       0.504143      1.395151

min          0.000000           0.000000       0.000000      0.000000

25%          0.000000           0.000000       0.000000      0.000000

50%          0.000000           0.000000       0.000000      0.000000

75%          0.000000           0.000000       0.420000      0.000000

max          4.540000          14.280000       5.100000     42.810000


       word_freq_our  word_freq_over  word_freq_remove
word_freq_internet  \
count    4601.000000     4601.000000       4601.000000
4601.000000
mean        0.312223        0.095901          0.114208
0.105295
std         0.672513        0.273824          0.391441
0.401071
min         0.000000        0.000000          0.000000
0.000000
25%         0.000000        0.000000          0.000000
0.000000
```

```
50%          0.000000        0.000000          0.000000
0.000000
75%          0.380000        0.000000          0.000000
0.000000
max         10.000000        5.880000          7.270000
11.110000

       word_freq_order  word_freq_mail  ...  char_freq_%3B  char_freq_
%28  \
count      4601.000000     4601.000000  ...    4601.000000
4601.000000
mean          0.090067        0.239413  ...       0.038575
0.139030
std           0.278616        0.644755  ...       0.243471
0.270355
min           0.000000        0.000000  ...       0.000000
0.000000
25%           0.000000        0.000000  ...       0.000000
0.000000
50%           0.000000        0.000000  ...       0.000000
0.065000
75%           0.000000        0.160000  ...       0.000000
0.188000
max           5.260000       18.180000  ...       4.385000
9.752000

       char_freq_%5B  char_freq_%21  char_freq_%24  char_freq_%23  \
count    4601.000000    4601.000000    4601.000000    4601.000000
mean        0.016976       0.269071       0.075811       0.044238
std         0.109394       0.815672       0.245882       0.429342
min         0.000000       0.000000       0.000000       0.000000
25%         0.000000       0.000000       0.000000       0.000000
50%         0.000000       0.000000       0.000000       0.000000
75%         0.000000       0.315000       0.052000       0.000000
max         4.081000      32.478000       6.003000      19.829000

       capital_run_length_average  capital_run_length_longest  \
count                 4601.000000                 4601.000000
mean                    5.191515                   52.172789
std                    31.729449                  194.891310
min                     1.000000                    1.000000
25%                     1.588000                    6.000000
50%                     2.276000                   15.000000
75%                     3.706000                   43.000000
max                  1102.500000                 9989.000000

       capital_run_length_total        class
count               4601.000000  4601.000000
mean                 283.289285     0.394045
std                  606.347851     0.488698
```

```
min                     1.000000     0.000000
25%                    35.000000     0.000000
50%                    95.000000     0.000000
75%                   266.000000     1.000000
max                 15841.000000     1.000000

[8 rows x 58 columns]
```

```
print(df.isnull().sum())
```

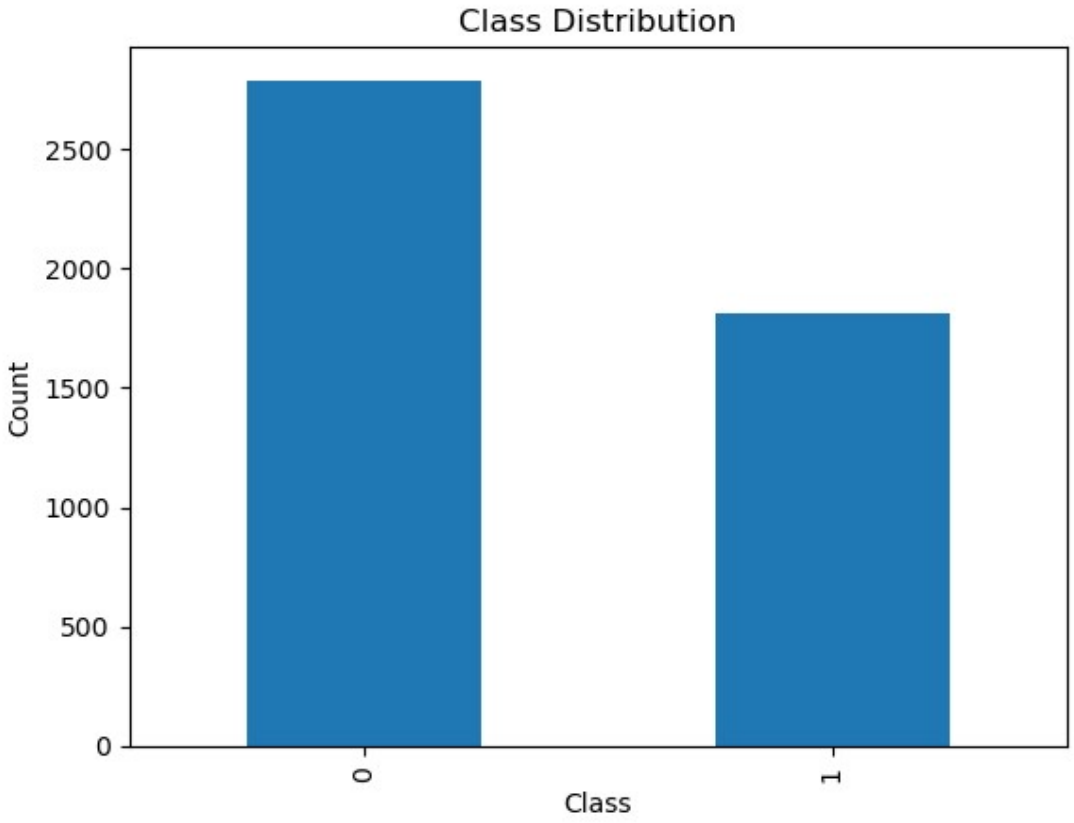
```
word_freq_make                0
word_freq_address             0
word_freq_all                 0
word_freq_3d                  0
word_freq_our                 0
word_freq_over                0
word_freq_remove              0
word_freq_internet            0
word_freq_order               0
word_freq_mail                0
word_freq_receive             0
word_freq_will                0
word_freq_people              0
word_freq_report              0
word_freq_addresses           0
word_freq_free                0
word_freq_business            0
word_freq_email               0
word_freq_you                 0
word_freq_credit              0
word_freq_your                0
word_freq_font                0
word_freq_000                 0
word_freq_money               0
word_freq_hp                  0
word_freq_hpl                 0
word_freq_george              0
word_freq_650                 0
word_freq_lab                 0
word_freq_labs                0
word_freq_telnet              0
word_freq_857                 0
word_freq_data                0
word_freq_415                 0
word_freq_85                  0
word_freq_technology          0
word_freq_1999                0
word_freq_parts               0
word_freq_pm                  0
word_freq_direct              0
```

```
word_freq_cs                    0
word_freq_meeting               0
word_freq_original              0
word_freq_project               0
word_freq_re                    0
word_freq_edu                   0
word_freq_table                 0
word_freq_conference            0
char_freq_%3B                   0
char_freq_%28                   0
char_freq_%5B                   0
char_freq_%21                   0
char_freq_%24                   0
char_freq_%23                   0
capital_run_length_average      0
capital_run_length_longest      0
capital_run_length_total        0
class                           0
dtype: int64
```

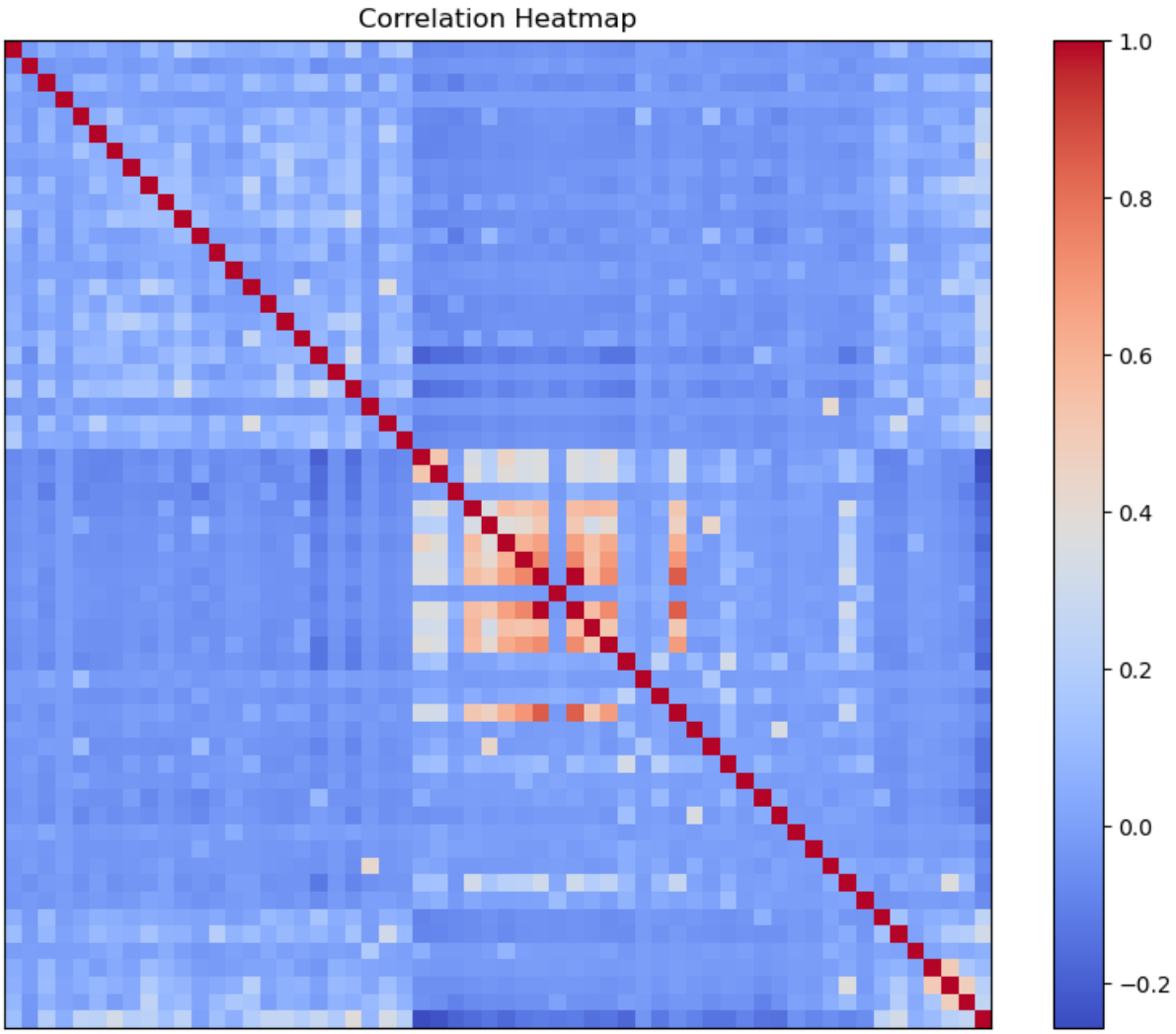
```
print("Duplicate Rows :", df.duplicated().sum())
```

```
Duplicate Rows : 391
```

```
df["class"].value_counts().plot(kind="bar")

plt.title("Class Distribution")
plt.xlabel("Class")
plt.ylabel("Count")

plt.show()
```

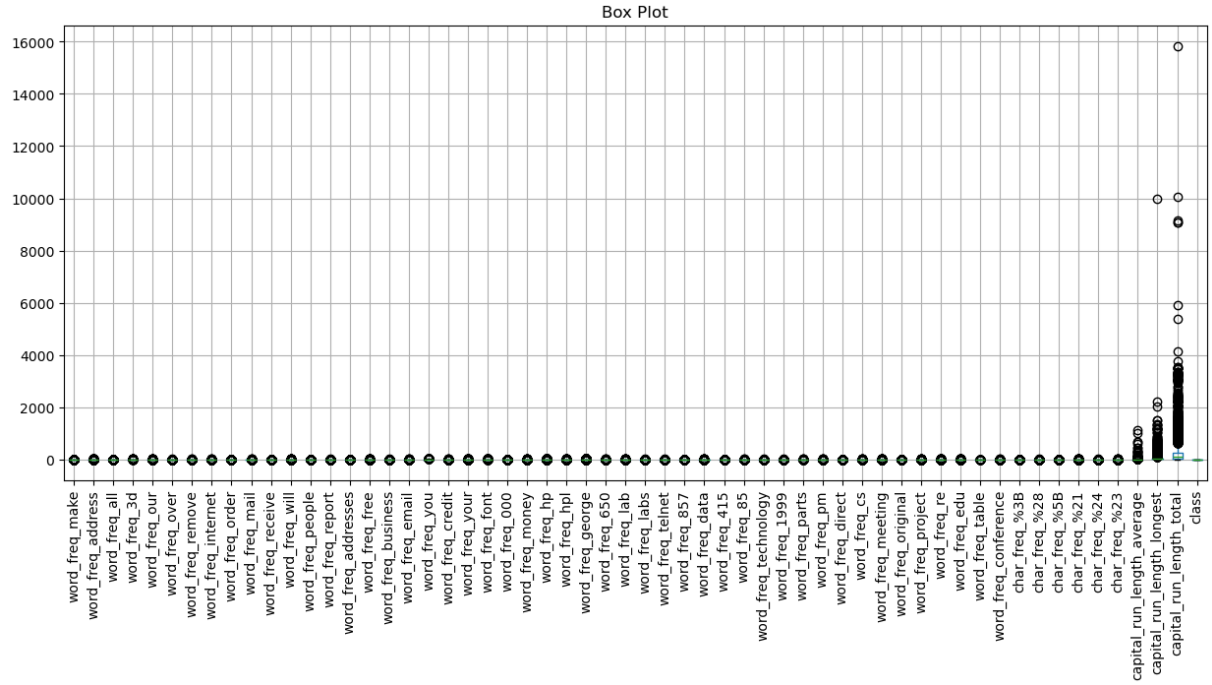



```
plt.figure(figsize=(10,8))

corr = df.corr()

plt.imshow(corr, cmap="coolwarm")

plt.colorbar()

plt.xticks([])
plt.yticks([])

plt.title("Correlation Heatmap")

plt.show()
```




```
df.hist(figsize=(15,12))

plt.tight_layout()

plt.show()
```


```
plt.figure(figsize=(15,6))

df.boxplot(rot=90)

plt.title("Box Plot")

plt.show()
```




```
X = df.drop("class", axis=1)
y = df["class"]

scaler = MinMaxScaler()

X = scaler.fit_transform(X)

print("Features Normalized")
```

```
Features Normalized
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)

print("Training Samples :", len(X_train))
print("Testing Samples :", len(X_test))
```

```
Training Samples : 3680
Testing Samples : 921
```

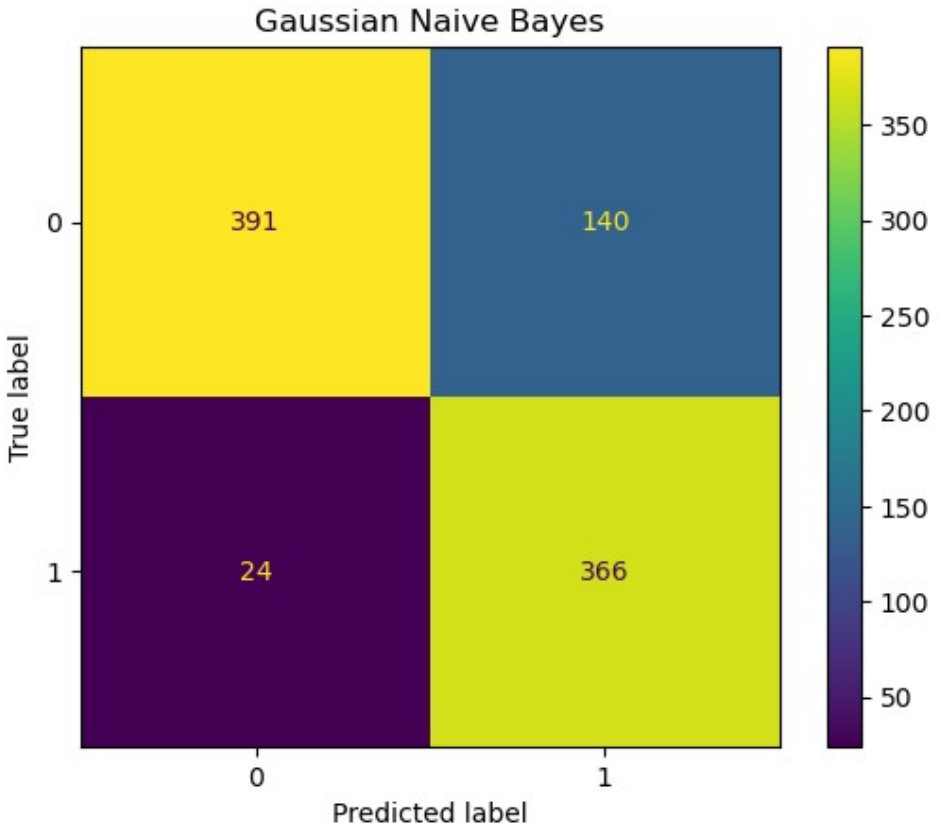
```
gnb = GaussianNB()

start = time.time()

gnb.fit(X_train, y_train)
```

```
train_time_gnb = time.time() - start
start = time.time()
y_pred_gnb = gnb.predict(X_test)
prediction_time_gnb = time.time() - start
mnb = MultinomialNB()
start = time.time()
mnb.fit(X_train, y_train)
train_time_mnb = time.time() - start
start = time.time()
y_pred_mnb = mnb.predict(X_test)
prediction_time_mnb = time.time() - start
bnb = BernoulliNB()
start = time.time()
bnb.fit(X_train, y_train)
train_time_bnb = time.time() - start
start = time.time()
y_pred_bnb = bnb.predict(X_test)
prediction_time_bnb = time.time() - start
knn = KNeighborsClassifier(n_neighbors=5)
start = time.time()
knn.fit(X_train, y_train)
train_time_knn = time.time() - start
start = time.time()
y_pred_knn = knn.predict(X_test)
prediction_time_knn = time.time() - start
```

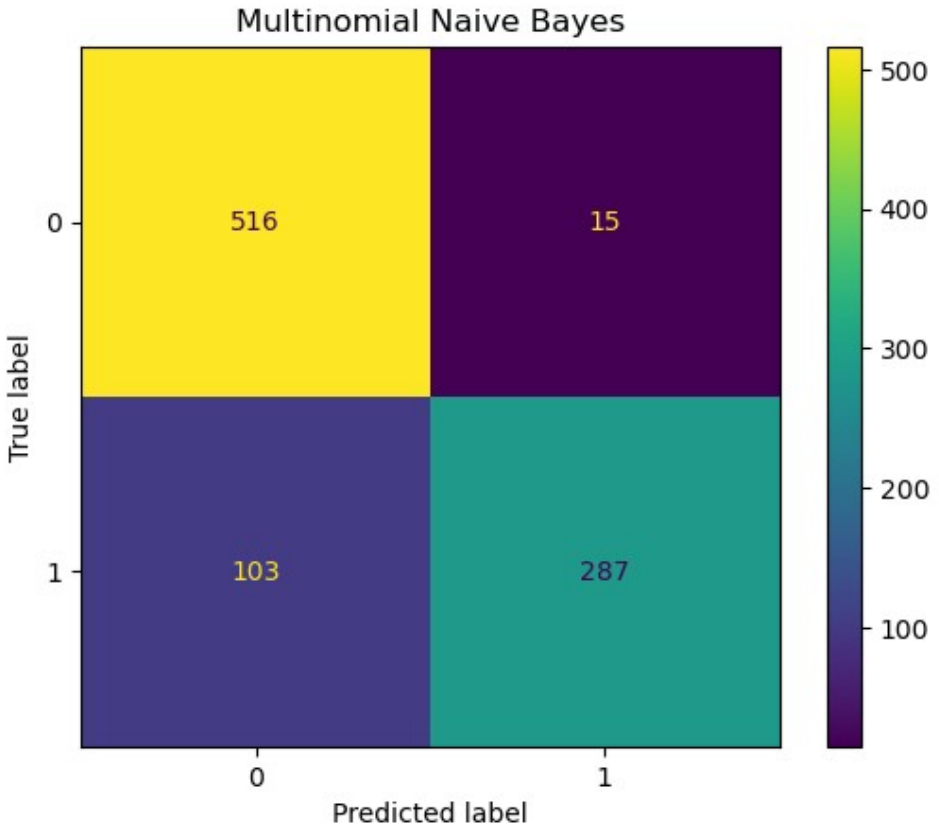
```
def evaluate(y_true, y_pred):

    accuracy = accuracy_score(y_true, y_pred)
    precision = precision_score(y_true, y_pred)
    recall = recall_score(y_true, y_pred)
    f1 = f1_score(y_true, y_pred)
    roc = roc_auc_score(y_true, y_pred)

    return accuracy, precision, recall, f1, roc

g = evaluate(y_test, y_pred_gnb)
m = evaluate(y_test, y_pred_mnb)
b = evaluate(y_test, y_pred_bnb)

nb_results = pd.DataFrame({
    "Metric": ["Accuracy", "Precision", "Recall", "F1", "ROC-AUC"],
    "Gaussian": g,
    "Multinomial": m,
    "Bernoulli": b
})

nb_results
```

	Metric	Gaussian	Multinomial	Bernoulli
0	Accuracy	0.821933	0.871878	0.880565
1	Precision	0.723320	0.950331	0.904624
2	Recall	0.938462	0.735897	0.802564
3	F1	0.816964	0.829480	0.850543
4	ROC-AUC	0.837404	0.853824	0.870209

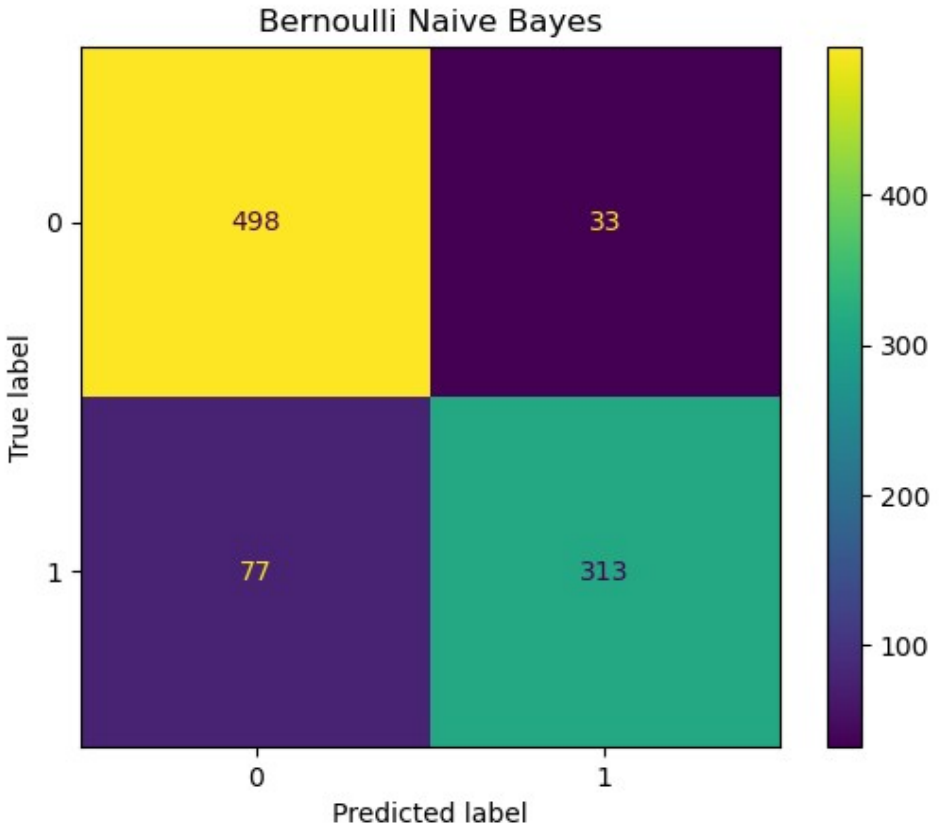
```
ConfusionMatrixDisplay.from_predictions(y_test, y_pred_gnb)

plt.title("Gaussian Naive Bayes")

plt.show()
```



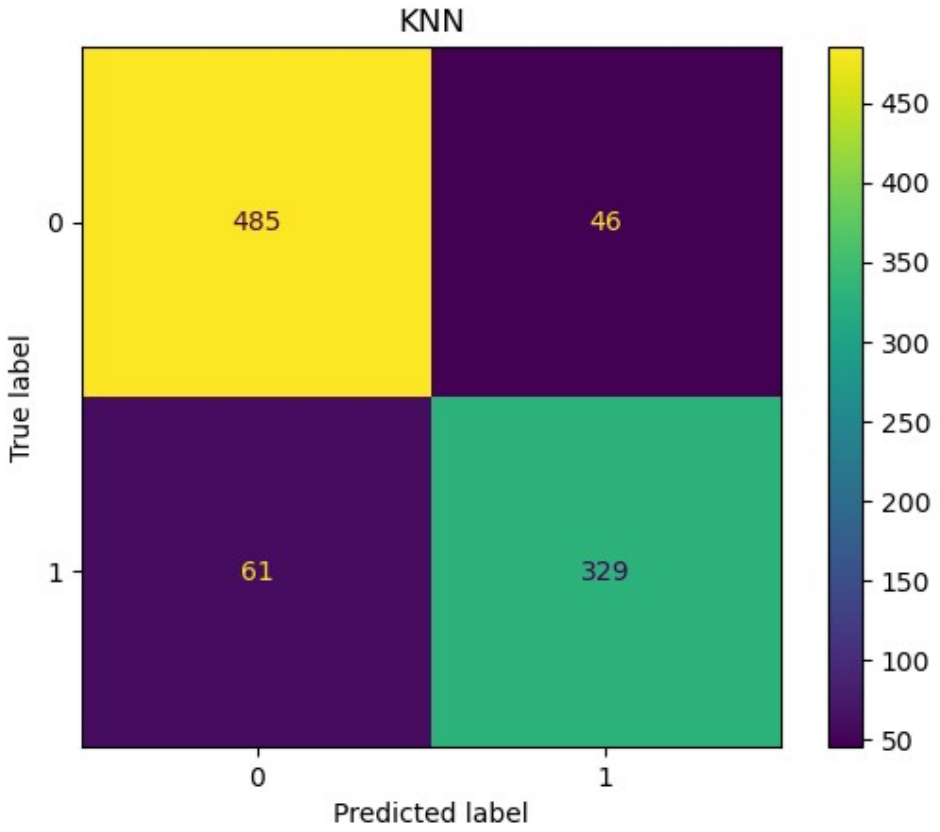

```
ConfusionMatrixDisplay.from_predictions(y_test, y_pred_mnb)

plt.title("Multinomial Naive Bayes")

plt.show()
```




```
ConfusionMatrixDisplay.from_predictions(y_test, y_pred_bnb)
plt.title("Bernoulli Naive Bayes")
plt.show()
```




```
ConfusionMatrixDisplay.from_predictions(y_test, y_pred_knn)

plt.title("KNN")

plt.show()
```

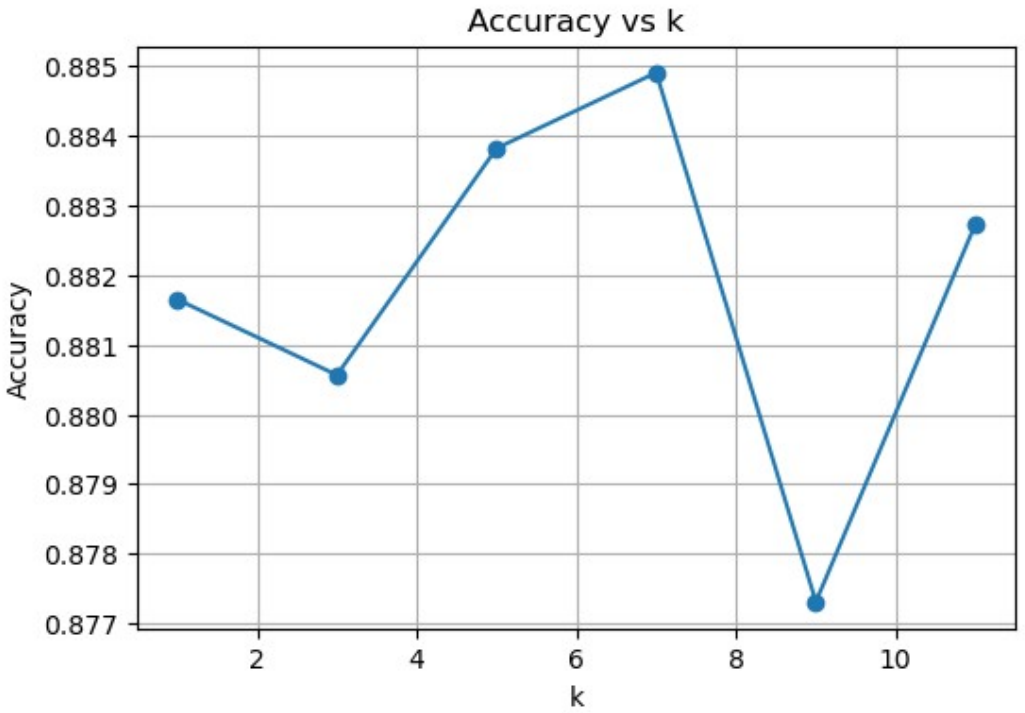



```
k_values = [1, 3, 5, 7, 9, 11]

results = []

for k in k_values:

    model = KNeighborsClassifier(n_neighbors=k)

    model.fit(X_train, y_train)

    pred = model.predict(X_test)

    accuracy = accuracy_score(y_test, pred)
    precision = precision_score(y_test, pred)
    recall = recall_score(y_test, pred)
    f1 = f1_score(y_test, pred)

    results.append([k, accuracy, precision, recall, f1])

knn_table = pd.DataFrame(
    results,
    columns=["k", "Accuracy", "Precision", "Recall", "F1"]
)
```

```
knn_table
```

	k	Accuracy	Precision	Recall	F1
0	1	0.881650	0.850374	0.874359	0.862200
1	3	0.880565	0.872340	0.841026	0.856397
2	5	0.883822	0.877333	0.843590	0.860131
3	7	0.884908	0.879679	0.843590	0.861257
4	9	0.877307	0.887955	0.812821	0.848728
5	11	0.882736	0.896067	0.817949	0.855228

```
plt.figure(figsize=(6,4))

plt.plot(
    knn_table["k"],
    knn_table["Accuracy"],
    marker="o"
)

plt.title("Accuracy vs k")

plt.xlabel("k")

plt.ylabel("Accuracy")

plt.grid(True)

plt.show()
```




```
parameters = {
    "n_neighbors": [1, 3, 5, 7, 9, 11],
    "weights": ["uniform", "distance"],
    "algorithm": ["auto", "kd_tree", "ball_tree"]
}

start = time.time()

grid_search = GridSearchCV(
    KNeighborsClassifier(),
    parameters,
    cv=5
)

grid_search.fit(X_train, y_train)

grid_time = time.time() - start

print("Best Parameters")
print(grid_search.best_params_)

print("\nBest CV Score")
print(grid_search.best_score_)
```

```
Best Parameters
{'algorithm': 'auto', 'n_neighbors': 5, 'weights': 'distance'}
```

```
Best CV Score
0.9141304347826086
```

```
random_search = RandomizedSearchCV(
    KNeighborsClassifier(),
    param_distributions=parameters,
    n_iter=10,
    cv=5,
    random_state=42
)

start = time.time()

random_search.fit(X_train, y_train)

random_time = time.time() - start

print("Best Parameters")
print(random_search.best_params_)

print("\nBest CV Score")
print(random_search.best_score_)
```

```
Best Parameters
{'weights': 'distance', 'n_neighbors': 5, 'algorithm': 'kd_tree'}

Best CV Score
0.9141304347826086
```

```
search_result = pd.DataFrame({
    "Parameter": [
        "Best k",
        "Weights",
        "Algorithm",
        "CV Accuracy",
        "Execution Time"
    ],
    "GridSearchCV": [
        grid_search.best_params_["n_neighbors"],
        grid_search.best_params_["weights"],
        grid_search.best_params_["algorithm"],
        grid_search.best_score_,
        grid_time
    ],
    "RandomizedSearchCV": [
        random_search.best_params_["n_neighbors"],
        random_search.best_params_["weights"],
        random_search.best_params_["algorithm"],
        random_search.best_score_,
        random_time
    ]
```

```
})

search_result
```

```
        Parameter GridSearchCV RandomizedSearchCV
0          Best k            5                   5
1         Weights     distance           distance
2       Algorithm         auto            kd_tree
3     CV Accuracy      0.91413            0.91413
4  Execution Time       28.398              9.394
```

```
algorithms = ["kd_tree", "ball_tree"]

tree_result = []

for algo in algorithms:

    model = KNeighborsClassifier(
        n_neighbors=5,
        algorithm=algo
    )

    start = time.time()
    model.fit(X_train, y_train)
    train_time = time.time() - start

    start = time.time()
    pred = model.predict(X_test)
    predict_time = time.time() - start

    accuracy = accuracy_score(y_test, pred)

    tree_result.append(
        [algo, accuracy, train_time, predict_time]
    )

tree_table = pd.DataFrame(
    tree_result,
    columns=[
        "Algorithm",
        "Accuracy",
        "Training Time",
        "Prediction Time"
    ]
)

tree_table
```

```
   Algorithm  Accuracy  Training Time  Prediction Time
0    kd_tree  0.883822          0.041            0.264
1  ball_tree  0.883822          0.020            0.229
```

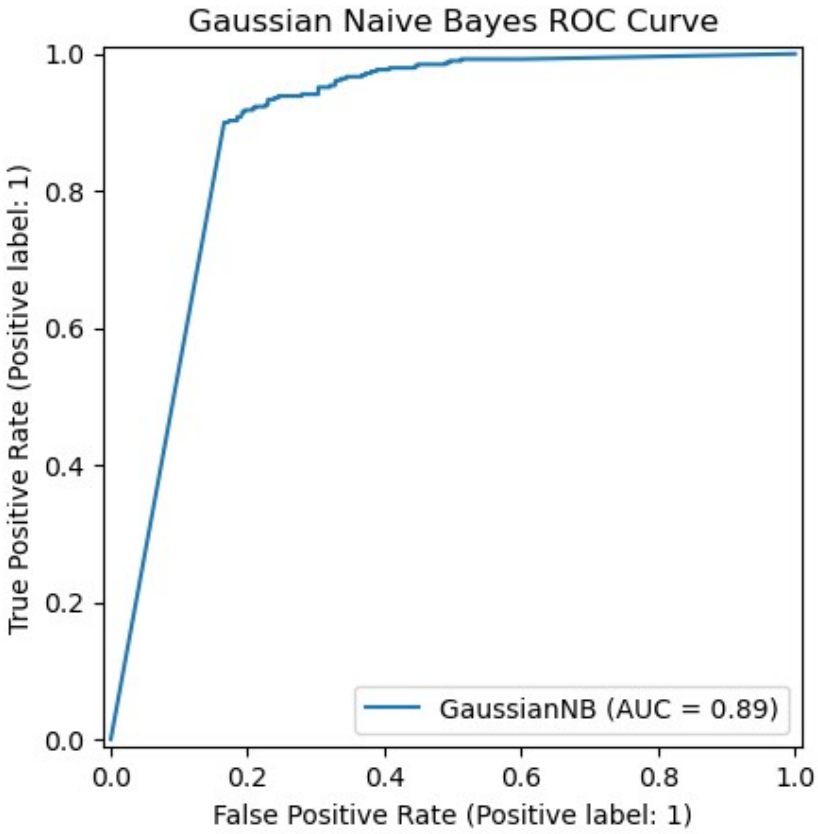
```
nb_cv = cross_val_score(GaussianNB(), X, y, cv=5)

best_knn = KNeighborsClassifier(
    n_neighbors=grid_search.best_params_["n_neighbors"],
    weights=grid_search.best_params_["weights"],
    algorithm=grid_search.best_params_["algorithm"]
)

knn_cv = cross_val_score(best_knn, X, y, cv=5)

cv_table = pd.DataFrame({
    "Fold":[1,2,3,4,5],
    "Naive Bayes":nb_cv,
    "Best KNN":knn_cv
})

cv_table
```

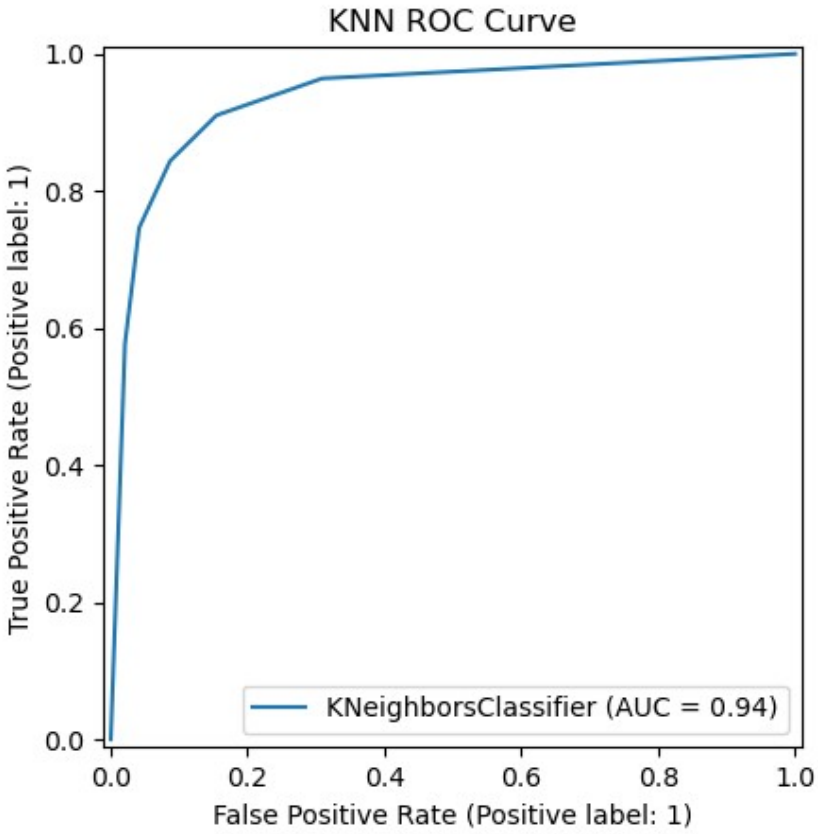
	Fold	Naive Bayes	Best KNN
0	1	0.851249	0.876221
1	2	0.866304	0.892391
2	3	0.854348	0.918478
3	4	0.843478	0.909783
4	5	0.695652	0.763043

```
average = pd.DataFrame({
    "Model":["Naive Bayes","Best KNN"],
    "Average Accuracy":[
        nb_cv.mean(),
        knn_cv.mean()
    ]
})

average
```

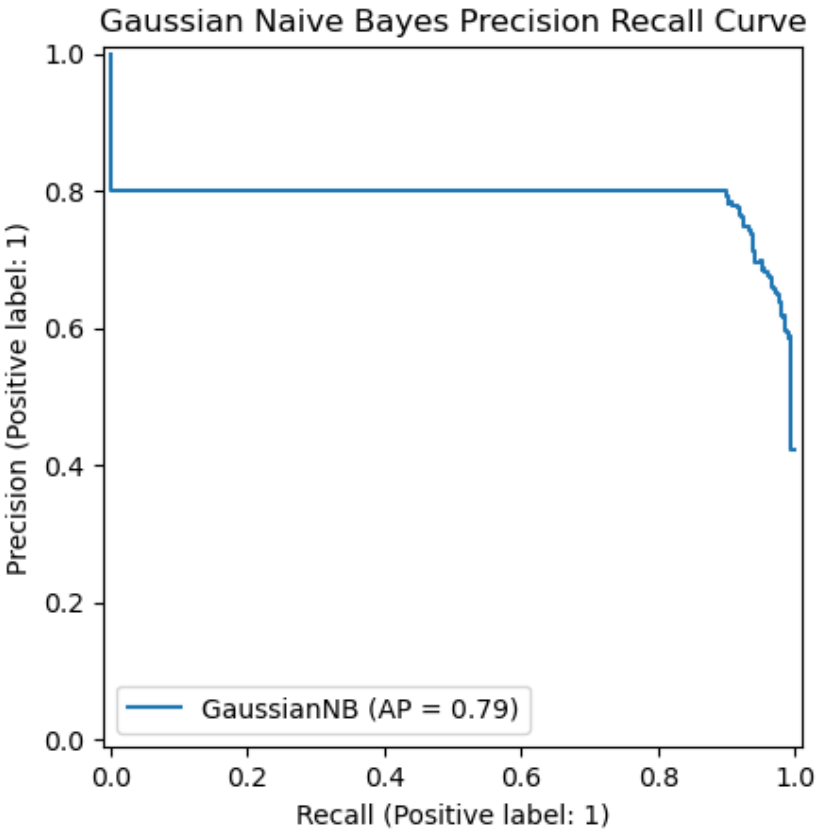
	Model	Average Accuracy
0	Naive Bayes	0.822206
1	Best KNN	0.871983

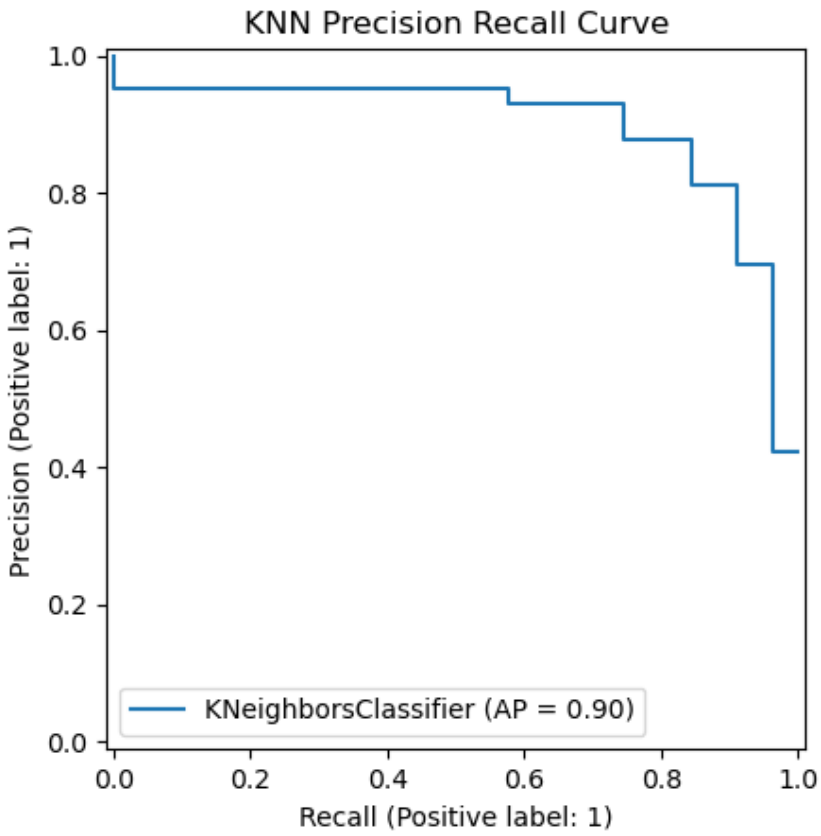
```
RocCurveDisplay.from_estimator(gnb, X_test, y_test)
plt.title("Gaussian Naive Bayes ROC Curve")
plt.show()

RocCurveDisplay.from_estimator(knn, X_test, y_test)
plt.title("KNN ROC Curve")
plt.show()
```


Gaussian Naive Bayes ROC Curve
True Positive Rate (Positive label: 1)
False Positive Rate (Positive label: 1)
GaussianNB (AUC = 0.89)
0.0
0.2
0.4
0.6
0.8
1.0




```
PrecisionRecallDisplay.from_estimator(gnb, X_test, y_test)
plt.title("Gaussian Naive Bayes Precision Recall Curve")
plt.show()

PrecisionRecallDisplay.from_estimator(knn, X_test, y_test)
plt.title("KNN Precision Recall Curve")
plt.show()
```


Gaussian Naive Bayes Precision Recall Curve
Precision (Positive label: 1)
Recall (Positive label: 1)
GaussianNB (AP = 0.79)
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0




```
training = pd.DataFrame({
    "Algorithm":[
        "Gaussian NB",
        "Multinomial NB",
        "Bernoulli NB",
        "Best KNN"
    ],
    "Training Time":[
        train_time_gnb,
        train_time_mnb,
        train_time_bnb,
        train_time_knn
    ]
})

training
```

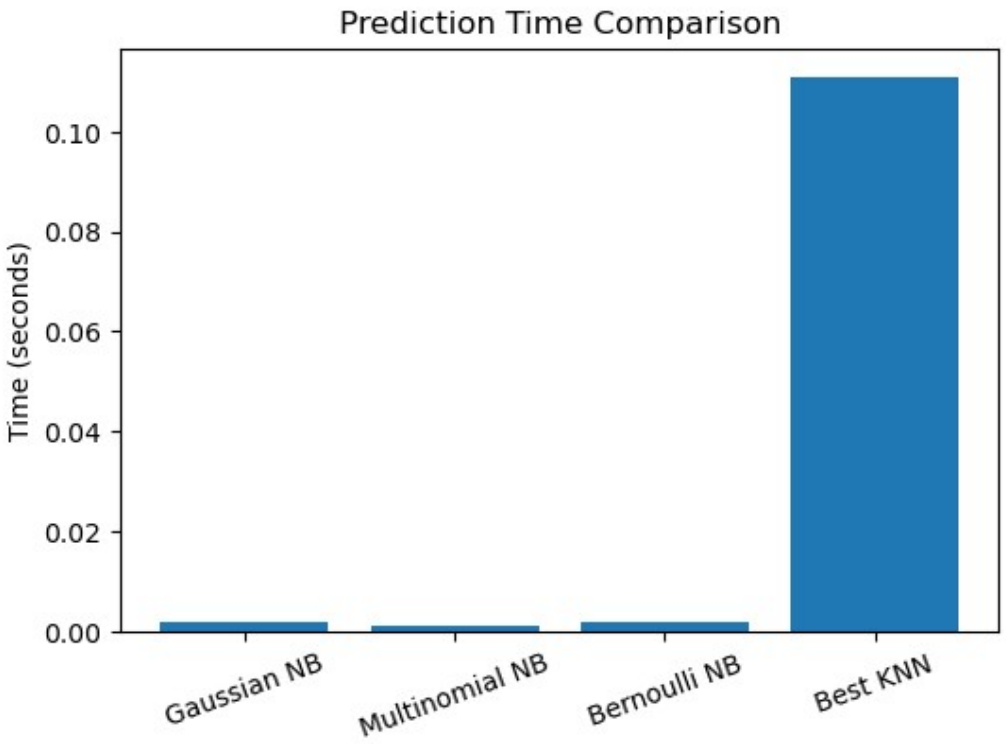
```
        Algorithm  Training Time
0     Gaussian NB          0.008
1  Multinomial NB          0.007
2    Bernoulli NB          0.012
3        Best KNN          0.002
```

```
prediction = pd.DataFrame({
    "Algorithm":[
        "Gaussian NB",
        "Multinomial NB",
        "Bernoulli NB",
        "Best KNN"
    ],
    "Prediction Time":[
        prediction_time_gnb,
        prediction_time_mnb,
        prediction_time_bnb,
        prediction_time_knn
    ]
})

prediction
```

```
        Algorithm  Prediction Time
0     Gaussian NB            0.002
1  Multinomial NB            0.001
2    Bernoulli NB            0.002
3        Best KNN            0.111
```

```
plt.figure(figsize=(6,4))

plt.bar(training["Algorithm"], training["Training Time"])

plt.title("Training Time Comparison")

plt.ylabel("Time (seconds)")

plt.xticks(rotation=20)

plt.show()
```


```
plt.figure(figsize=(6,4))
plt.bar(prediction["Algorithm"], prediction["Prediction Time"])
plt.title("Prediction Time Comparison")
plt.ylabel("Time (seconds)")
plt.xticks(rotation=20)
plt.show()
```

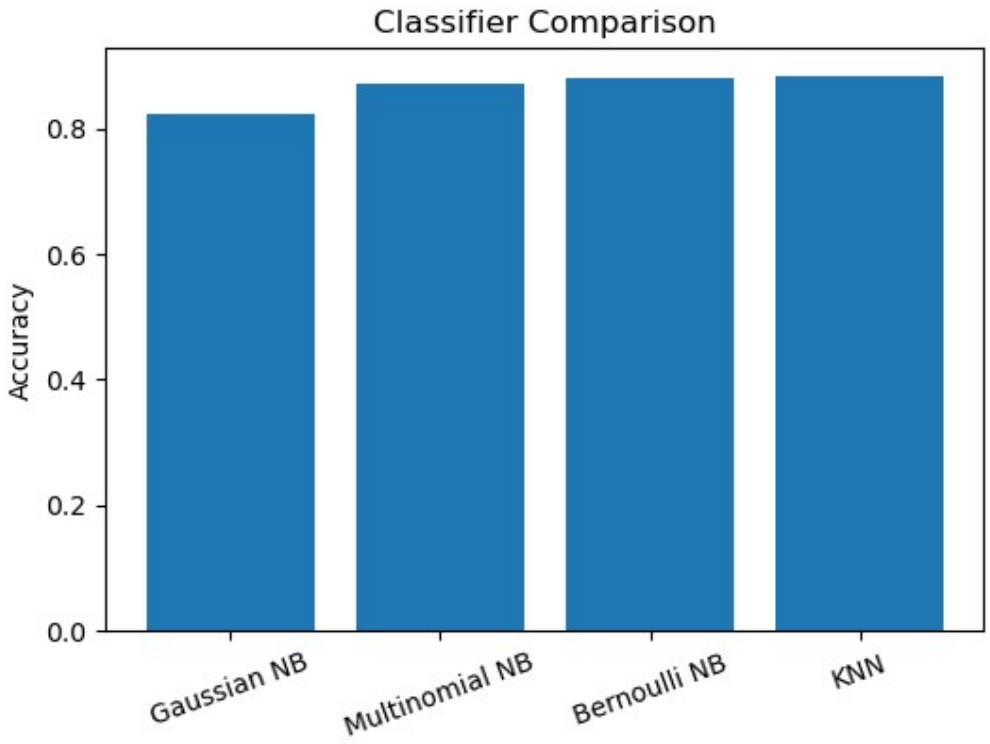



```
classifier = pd.DataFrame({
    "Classifier":[
        "Gaussian NB",
        "Multinomial NB",
        "Bernoulli NB",
        "KNN"
    ],
    "Accuracy":[
        g[0],
        m[0],
        b[0],
        evaluate(y_test, y_pred_knn)[0]
    ]
})

classifier
```

```
       Classifier  Accuracy
0     Gaussian NB  0.821933
1  Multinomial NB  0.871878
2    Bernoulli NB  0.880565
3             KNN  0.883822
```

```
plt.figure(figsize=(6,4))

plt.bar(classifier["Classifier"], classifier["Accuracy"])
```

```
plt.title("Classifier Comparison")
plt.ylabel("Accuracy")
plt.xticks(rotation=20)
plt.show()
```



Hyperparameter Tuning

GridSearchCV and RandomizedSearchCV were used to find suitable parameters for the KNN classifier. Different values of k, weights and algorithms were tested and the best combination was selected based on cross-validation accuracy.

Comparison Tables

Comparison tables for Naïve Bayes models, KNN models, GridSearchCV, RandomizedSearchCV, KDTree, BallTree and Cross Validation are included in the attached notebook.

Complexity Analysis

Algorithm	**Training**	**Prediction**
Naïve Bayes	O(nd)	O(d)
KNN	O(1)	O(nd)
KDTree	O(n log n)	O(log n)
BallTree	O(n log n)	O(log n)

Plots

The experiment includes the following plots.

- Class distribution
- Correlation heatmap
- Histograms
- Boxplots
- Confusion matrices
- ROC curves
- Precision-Recall curves
- Accuracy vs k
- Cross-validation accuracy
- Training time comparison
- Prediction time comparison
- Classifier comparison bar chart

Results

The models were trained and evaluated successfully. Their performance was compared using accuracy, precision, recall, F1-score and ROC-AUC. The comparison helped identify the better model for email spam classification.

Conclusion

In this experiment, Email Spam/Ham classification was performed using Naïve Bayes and KNN. The performance of different models was compared using various evaluation metrics. Hyperparameter tuning and cross validation helped in selecting a suitable model for the given dataset.